

Capstone Project

Countries Ratings Prediction Tool Final Report

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1. Executive Summary

The **Countries Ratings Project** is designed to predict **sovereign credit ratings and outlooks** for the current year or any future year where macroeconomic forecasts are available. This capability is critical for companies, institutional investors, and individuals who need to evaluate new opportunities, compare investment attractiveness across countries, and assess associated risk levels.

The model is trained on **historical credit ratings** from S&P, Moody's, and DBRS, combined with **macroeconomic and demographic data** sourced from the World Bank and IMF. Predictions are aligned with the rating scales used by leading agencies such as Moody's, Fitch, and S&P.

The **Country Rating Prediction Tool** delivers on-demand ratings and outlooks for any selected country, providing investors with **timely, data-driven insights** into sovereign creditworthiness and investment risk.

Project GitHub Link: [Countries Ratings Tool](#)

2. Problem Statement

Currently, there are a few problems with the Country Ratings provided by the rating agencies:

1. Not all countries have official credit ratings, and this is an issue for investors that want to invest in countries that do not have country ratings.
2. Even when available, ratings are often updated infrequently. This is because credit ratings agencies release the updates based on a particular schedule which is not always frequent enough.

3. Model Outcomes or Predictions

This project is using a supervised machine learning algorithm with classification models. The classifications models are applied to two multiclass targets as defined below:

1. Country Rating, a 29 classes/values defined by the ratings between AAA and C as per country ratings agencies scales.
2. Country Outlook, which is defined by three classes/values (*Negative, Stable or Positive*).

For any chosen country, for a given number of macroeconomic indicators (45 indicators listed in the **Annex - A** below), the tool will predict the Country Rating and its Outlook.

4. Data Acquisition:

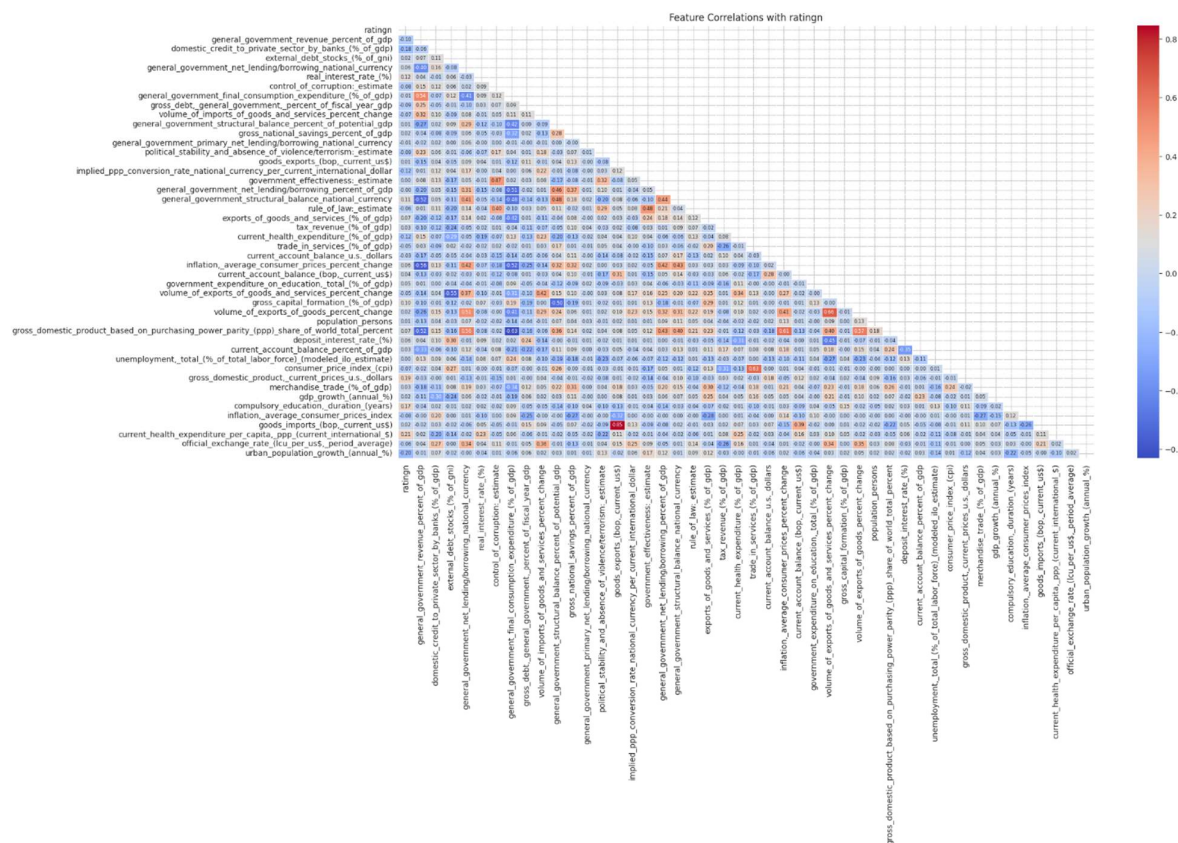
For this project I've used the following sources of information:

Dataset	Source
Countries Credit Ratings	Wikipedia
CPI (Inflation)	IMF
Government Debt	IMF
Macroeconomic Indicators	World Bank

The rationale behind the selected sources of information is given by the relationship between the macroeconomic, political, social and demographic indicators of a country and the overall country rating and its outlook.

Selected datasets are timeseries that contains historical information covering period **1960–2023**.

Below heatmap provides the correlation between inputs indicators and the Country Rating and Country Outlook.



5. Data Preprocessing/Preparation

Dataset loading and Transformation

I have used five different datasets, (*World Indicators, World Economic Indicators, Countries CPI, Governments Debt, and Countries Ratings*) from three sources: **IMF, World Bank and Wikipedia**.

Considering the diversity of the datasets and their different formats, I had to extract, filter, transform and merge all different datasets into one single dataset. To achieve this, I've used techniques like: clean duplicates, change datatypes, standardize names and transform columns.

For the historical countries' ratings, I had to consolidate all ratings from different ratings agencies into one normalized rating that then was mapped to numerical values. This provided 29 classes corresponding to all credit ratings currently in use.

I've applied the same technique for the countries outlook labels, which after preprocessing ended up with 3 classes (Negative, Stable or Positive).

All the above preprocessing techniques created a consolidate and clean dataset.

Exploratory Data Analysis (EDA)

Data Cleanup

- Removed all non-country entities
- Removed Countries that have less than three values in the country rating during the period
- Remove years period [1960 – 1980] and year [2024] due to missing data and keep only **period [1980 - 2023]**
- Removed categories that contains too many missing values or categories that are not needed
- Removed categories that shows multi-collinearity
- Analyzed and processed outliers

Imputation

Performed imputation on some missing target values using the following techniques: *Forward fill* – to fill in forward most of the missing values and *Backward fill* – to fill-in the missing values before the first available record

Performed imputation to address the missing values in features columns. I've used a staged imputation process that used one of the following methods: *polynomial fill, linear fill, rolling average fill, ARIMA and global fallback(mean fill)*.

Principal Component Analysis (PCA)

I've used PCA for dimensionality reduction and selected (first principal components). Performed Train/Test split using 20% of the samples for the test set. Used a preprocessor to encode numerical columns with StandardScaler encoder and categorical columns with OneHotEncoder. Selected only principal components that explains 90% of the data correlation and have a correlation of at least 0.01.

Addressed multiclass imbalance using oversampling techniques for both targets. I've Improved minority class imbalance using *Borderline SMOTE* in case of country ratings, and *ADASYN* in case of countries outlook.

6. Modeling:

I've defined the **Baseline Model accuracy** for using *Decision Tree Classifier (max-depth = 40)* and used **F1-Score** as the key metric. Being a multiclass classification F1-Score is a more accurate representation of the overall models' performances.

Below the reference F1-Score for each target:

Baseline results	F1-Score
Country Credit Ratings (yr) - DecisionTree	0.5261
Country Outlook (yo) - DecisionTree	0.5271

I've experimented with different supervised classification models but at the end I've chosen the following models: *RandomForest*, *XGBoost*, *HistGradientBoosting*, *ExtraTrees* and *LightGBM*. Other models like *CatBoost* or *MLPClassifier* (from SciKit Learn Neural Network) didn't perform well and decided to remove them from my experiments.

I've used both *GridSearchCV* and *RandomizedGridSearchCV* (selected for faster results) to conduct hyperparameter tuning, select the best models and best parameters. I've also selected top 20 most important features for the best model. For the best model I also list the best hyperparameters.

7. Models Evaluation

7.1. Countries Ratings model

For Country Ratings, **RandomForest** model provided the best results with an **F1-Score** of **71.10%** as an average across all evaluation metrics. Also, the training time was very good for this model when compared with the others. Unexpectedly, *LightGBM* model was the most computationally expensive.

There is some overfitting present in some models, however this could be handled by further finetuning on the selected model at a later stage. Below the final results of the experiments.

```
=====
Multiclass Model Results
=====
```

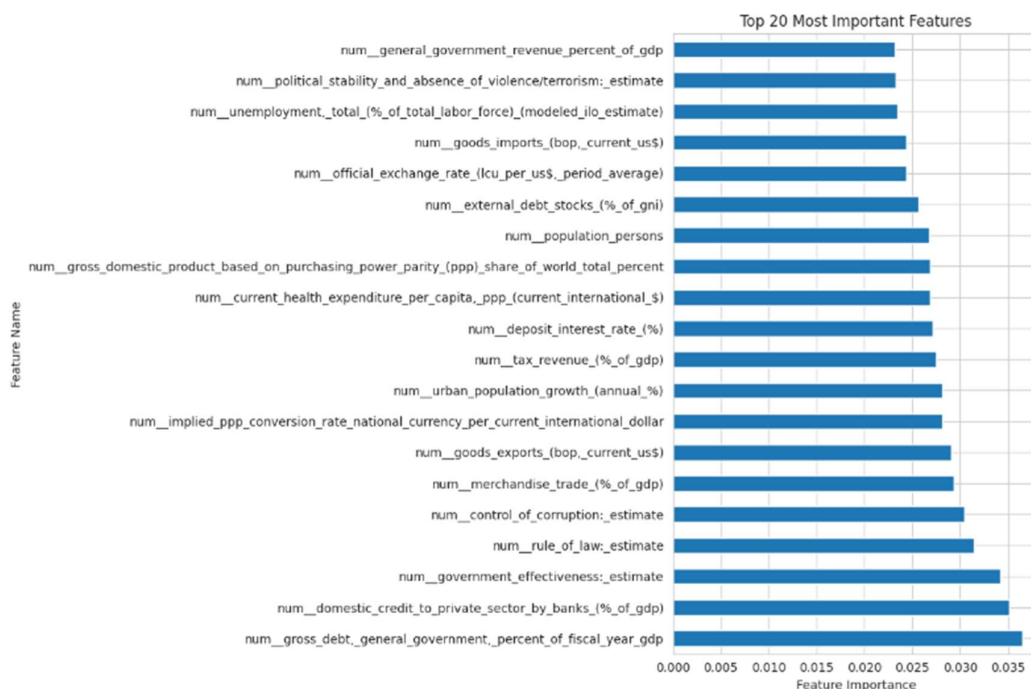
model	Model Performance Metrics						average fit time
	train score	test score	accuracy	recall	precision	f1_score	
randomforest	100.00%	77.91%	77.91%	72.37%	73.25%	71.10%	6.34s
xgboost	99.98%	77.18%	77.18%	70.65%	70.20%	69.57%	11.36s
histgradientboosting	98.59%	75.91%	75.91%	68.98%	69.03%	68.25%	24.22s
extratrees	99.39%	77.18%	77.18%	70.46%	67.65%	68.66%	1.94s
lightgbm	99.68%	76.00%	76.00%	69.80%	69.96%	69.09%	121.56s

```
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```

Below I have listed the best parameters for the best model:

```
=====
Overall Best Model and Parameters
=====
Best Model: randomforest
Best Test Score: 77.91%
Best Parameters: {'randomforest__n_estimators': 100, 'randomforest__min_samples_split': 2, 'randomforest__min_samples_leaf': 1, 'randomforest__max_depth': 25}
=====
```

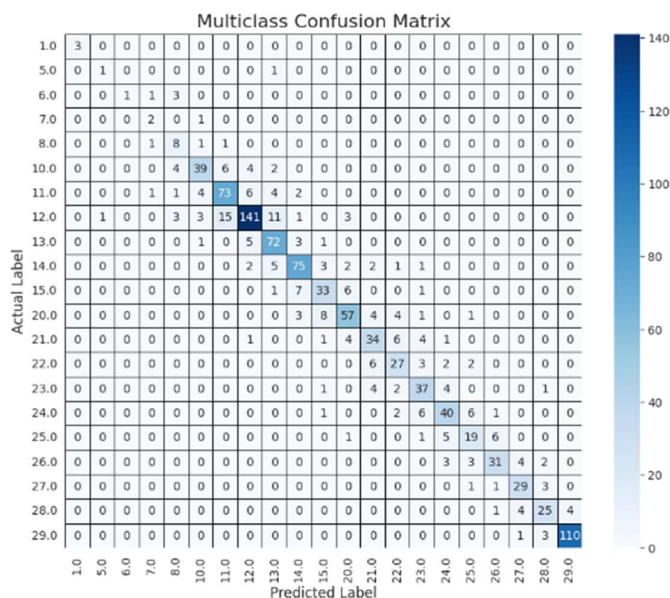
Top 20 features by importance for the best model **RandomForest** are provided below. This list provides an important business insight into what macroeconomic data should be considered as the most relevant to the final country ratings.



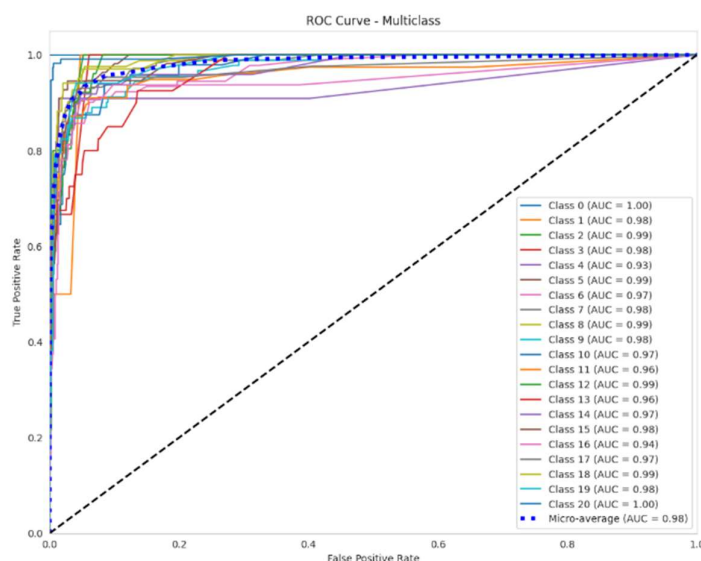
From the **Classification Report** we observe that minority classes have weaker results than the other classes. This indicates that there is additional room for improvement with regards to minority classes model performances. We can notice that for classes between 1 and 4 there are very few data samples therefore the misclassification is higher. Therefore, additional techniques that address unbalanced data challenges should be considered for further improvements. Alternatively, additional data concerning minority classes should be collected.

	precision	recall	f1-score	support
0	100.00%	100.00%	100.00%	3
1	50.00%	50.00%	50.00%	2
2	100.00%	20.00%	33.33%	5
3	40.00%	66.67%	50.00%	3
4	42.11%	72.73%	53.33%	11
5	79.59%	70.91%	75.00%	55
6	76.84%	80.22%	78.49%	91
7	88.68%	79.21%	83.68%	178
8	75.00%	87.80%	80.90%	82
9	82.42%	82.42%	82.42%	91
10	68.75%	68.75%	68.75%	48
11	78.08%	73.08%	75.50%	78
12	68.00%	66.67%	67.33%	51
13	64.29%	67.50%	65.85%	40
14	68.52%	75.51%	71.84%	49
15	72.73%	71.43%	72.07%	56
16	59.38%	59.38%	59.38%	32
17	77.50%	72.09%	74.70%	43
18	76.32%	85.29%	80.56%	34
19	73.53%	73.53%	73.53%	34
20	96.49%	96.49%	96.49%	114
accuracy	77.91%	77.91%	77.91%	1
macro avg	73.25%	72.37%	71.10%	1100
weighted avg	78.61%	77.91%	77.97%	1100

Below **Confusion Matrix** reveals that minority classes with very few samples are the ones that perform poorly. This indicates that more data is needed or additional oversampling might be required for an improved result for the minority classes.



Below ROC curves shows pretty good results for each class with few exceptions related to minority classes. Micro-Average AUC of 0.98 is a robust result.



Key messages for the business users related to the Countries Ratings:

Although overall, the model can predict with more than **71.10%** accuracy (using F1-Score) what is/will be the country rating, a more important information is given by the **Classification Report** that describes these scores by country rating classes. As can be observed in some cases the predictions per class are above and below this average value.

Considering that the cost of misclassification for the business is very high, the most important metric to be used as reference is **Recall score** (% of correctly identified classes). Therefore, the business user should check the recall scores for each class to understand how accurate the model can provide predictions for each class. Classification Report above provides a very clear and convenient way to check for this information. The mapping between model classes and different agencies ratings models is provided in **Annex - B** below.

Impact of the above predictions on investment's risks:

Assumptions:

Let's assume the investor is holding **10 million USD** in **10-year bonds** of a country that last time was rated as "**Investment grade (Lowest Tier)**" with any of the following ratings (Baa3/BBB/BBB or BBB(low)) meaning "**Class 11**" in our model.

The investor would need to know what will be the country rating today or in the near future to decide if they should exit or not at the optimal moment. Many Investment institutions are not allowed to hold "**Speculative Grade (Junk)**" bonds.

For this, the investment institution will need to know the expected new country rating. Will it move to “Speculative Grade (Junk): Highest Quality Junk” which in our model is “Class 10”? If this is happening the investor would need to exit the market before the downgrade. According to the model the recall accuracy for “Class 11” is 73.08% and for “Class 10” is 68.75%. In this case we shall consider the “**Class 10**” recall accuracy of **68.75%**. For reference, please check Annex – B.

The cost of country downgrade for the investor bond price

A country ratings downgrade causes a yield spread jump of about 150–300 basis points (1.5–3.0%).

Price change for a bond in this case is calculated with below formula:

Price change (%) ≈ -Bond Duration × ΔYield

Assuming that the bond duration left is only 7 years and the yield spread jump at maximum of 3% the expected bond price change will be:

-7 years x 3% = - **21%** of the bond value which in this case equals to

10 million x 21% = - **2.1 million USD**

Therefore, **the cost of missing the country downgrade prediction for the investor**, would be **2.1million**. When a downgrade to Class 10 actually happens, the model will correctly identify ~69% of cases.

The question to the business user is:

Given a potential \$2.1M loss per \$10M exposure, is a ~69% recall acceptable, or do we need to improve the model before using it in production risk monitoring?

7.2. Countries Outlook model

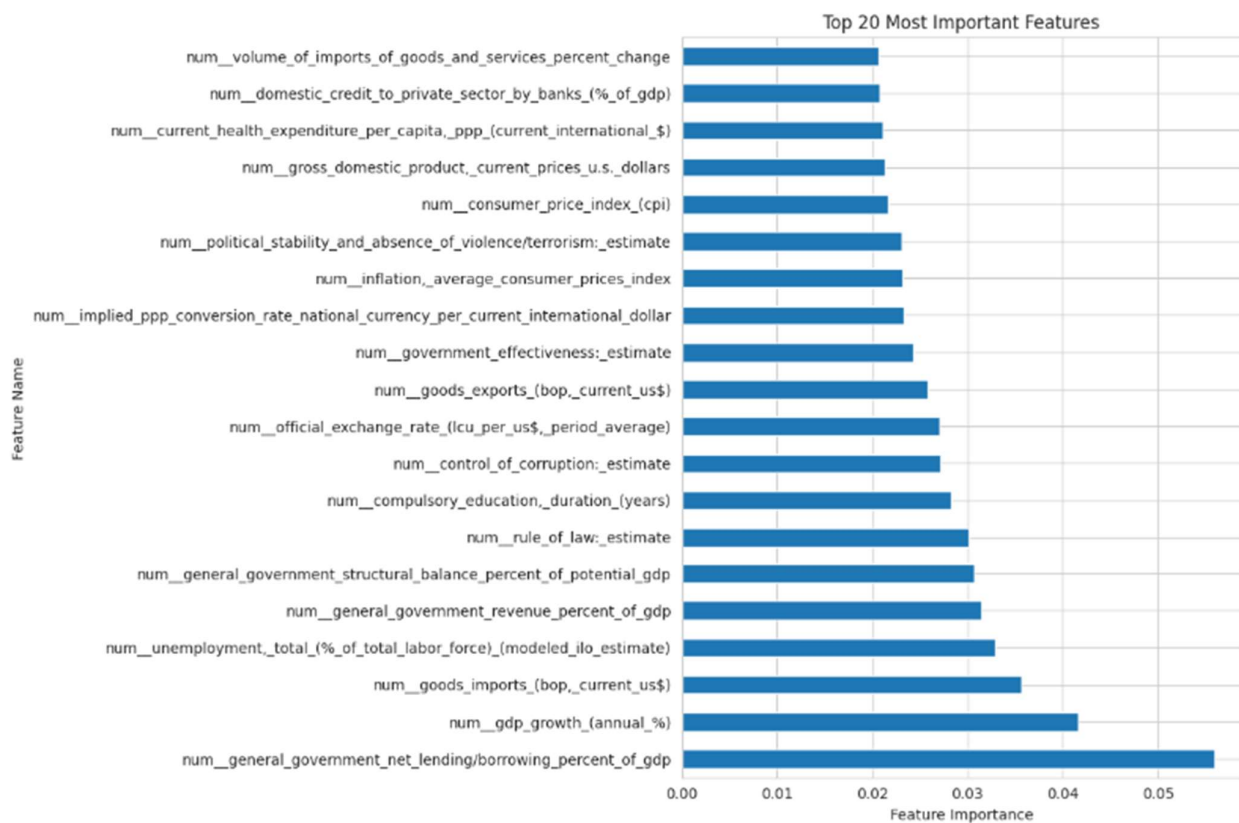
For Country Outlook, again **RandomForest** model with an **F1-Score** of **63.10%** followed by XGBoost model, provided the best results across all different evaluation metrics. Unexpectedly, again LightGBM model was the most computationally expensive. Also, in this case we can notice some overfitting. This should be handled by some further finetuning on the selected model.

Multiclass Model Results							
model	Model Performance Metrics						
	train score	test score	accuracy	recall	precision	f1_score	average fit time
randomforest	100.00%	76.91%	76.91%	65.92%	61.43%	63.10%	5.97s
xgboost	100.00%	78.36%	78.36%	61.26%	62.68%	61.94%	6.24s
histgradientboosting	98.64%	77.64%	77.64%	61.31%	61.53%	61.34%	2.16s
extratrees	77.61%	64.36%	64.36%	67.49%	53.48%	54.88%	2.13s
lightgbm	95.39%	75.45%	75.45%	63.05%	58.94%	60.65%	23.59s

Below I have listed the best parameters for RandomForest model:

```
=====
Overall Best Model and Parameters
=====
Best Model: xgboost
Best Test Score: 78.36%
Best Parameters: {'xgboost__subsample': 0.6, 'xgboost__n_estimators': 200, 'xgboost__min_child_weight': 2, 'xgboost__max_depth': 25, 'xgboost__learning_rate': 0.1, 'xgboost__colsample_bytree': 1}
```

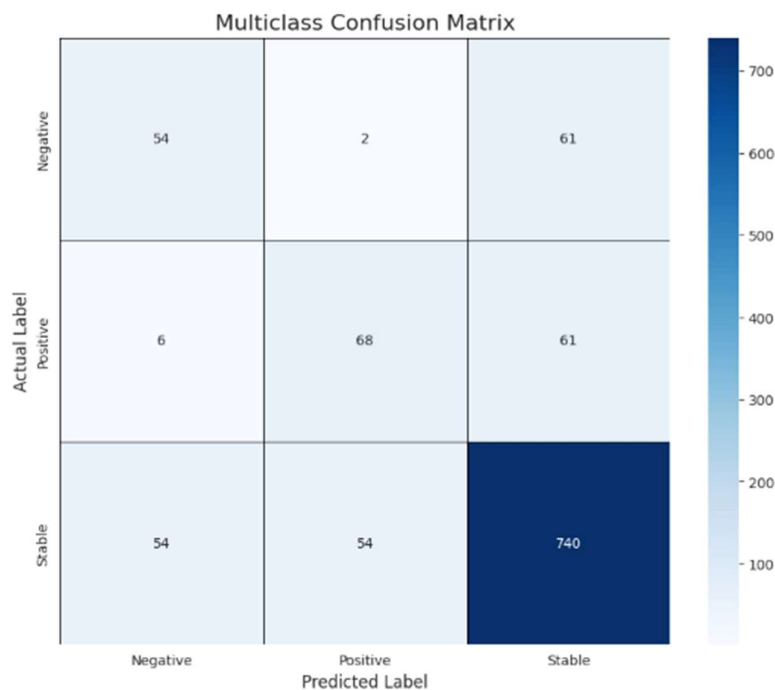
Top 20 features by importance for the best model, **RandomForest** are provided below. This list provides an important business insight into what macroeconomic data should be considered as the most relevant to the final country ratings.



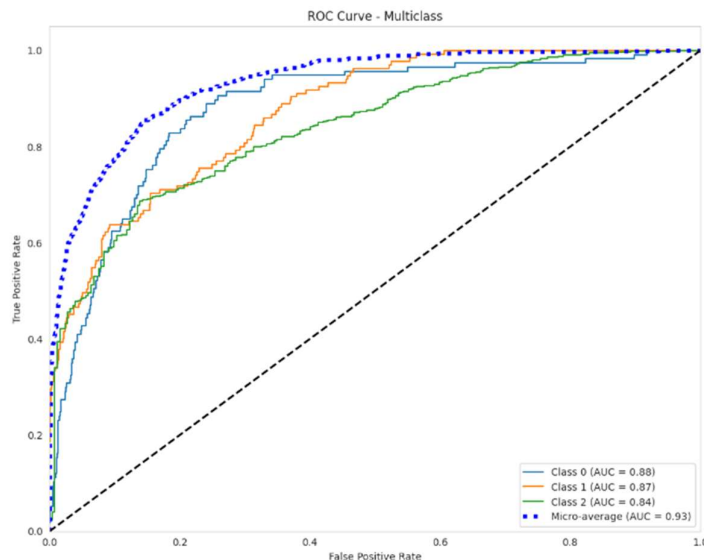
From the **Classification Report** we observe that minority classes (class 0 and 1) have significant weaker results than the other classes. This indicates that there is additional room for improvement with regards to minority classes models performances. Therefore, additional techniques that address unbalanced data challenges should be considered for further improvements. Alternatively, additional data concerning minority classes should be collected.

Classification Report				
	precision	recall	f1-score	support
0	47.37%	46.15%	46.75%	117
1	54.84%	50.37%	52.51%	135
2	85.85%	87.26%	86.55%	848
accuracy	78.36%	78.36%	78.36%	1
macro avg	62.68%	61.26%	61.94%	1100
weighted avg	77.95%	78.36%	78.14%	1100

Below **Confusion Matrix** reveals that minority classes with very fewer samples are the ones that perform poorly. We can notice that for classes 1 and 2, there are significant misclassifications. This indicates that more data is needed or additional oversampling might be required for an improved result for the minority classes.



Below **ROC curves** show modest results for each class. **Micro-Average AUC** of **0.93** is mainly driven by the majority class performance and is rather misleading in this case.



Key messages for the business users related to the Countries Outlook:

In this case the model can predict with more than **63%** accuracy (using F1-Score) what is/will be the country outlook. However, a more important information is given by the **Classification Report** that describes these scores by country outlook classes. As can be observed in some cases the predictions per class are above (for majority class) and below (for minority classes) accuracy average value.

Considering that the cost of misclassification for the business is very high, the most important metric to be used as reference is **Recall score** (% of correctly identified classes). Therefore, the business user should check the recall scores for each class to understand how accurate the model can provide predictions for each class.

Classification Report above provides a very clear and convenient way to check for this information.

It is noticeable that for “Stable Outlook” (class 2) the Recall Score is 87.27% which means that the model is accurate in over 87% of cases that are predicting a Stable Country Outlook.

However, the “Negative Outlook” (class 0) and “Positive Outlook” (class 1) classes perform poorly at 46.15% and respectively 50.37%. This requires further improvements (e.g. acquire more data, further address the minority class imbalance, explore other ML models) that will be taken as next steps.

ANNEX – A

#	INDICATOR	IMPORTANCE
1	gross_domestic_product_based_on_purchasing_power_parity(ppp)_share_of_world_total_percent	0.3328
2	general_government_final_consumption_expenditure_(%_of_gdp)	-0.3279
3	inflation_average_consumer_prices_percent_change	0.3030
4	general_government_revenue_percent_of_gdp	-0.2855
5	volume_of_exports_of_goods_percent_change	0.2848
6	general_government_structural_balance_national_currency	0.2799
7	general_government_net_lending/borrowing_national_currency	0.2757
8	general_government_net_lending/borrowing_percent_of_gdp	0.2673
9	volume_of_exports_of_goods_and_services_percent_change	0.2433
10	general_government_structural_balance_percent_of_potential_gdp	0.2361
11	merchandise_trade_(%_of_gdp)	0.1645
12	gross_national_savings_percent_of_gdp	0.1518
13	exports_of_goods_and_services_(%_of_gdp)	0.1477
14	current_account_balance_percent_of_gdp	0.1412
15	unemployment_total_(%_of_total_labor_force)(modeled_ilo_estimate)	-0.1367
16	external_debt_stocks_(%_of_gni)	-0.1039
17	gross_debt_general_government_percent_of_fiscal_year_gdp	-0.0916
18	population_persons	0.0883
19	rule_of_law_estimate	0.0829
20	gdp_growth(annual_%)	0.0755
21	goods_exports_(bop_current_us\$)	0.0700
22	deposit_interest_rate_(%)	-0.0667
23	real_interest_rate_(%)	-0.0602
24	official_exchange_rate_(lcu_per_us\$,period_average)	0.0580
25	control_of_corruption_estimate	-0.0560
26	domestic_credit_to_private_sector_by_banks_(%_of_gdp)	0.0472
27	current_account_balance_u.s._dollars	0.0468

28	gross_capital_formation_(%_of_gdp)	-0.0409
29	government_expenditure_on_education,_total_(%_of_gdp)	-0.0382
30	compulsory_education,_duration_(years)	-0.0380
31	urban_population_growth_(annual_%)	0.0365
32	current_account_balance_(bop,_current_us\$)	0.0334
33	consumer_price_index_(cpi)	0.0297
34	implied_ppp_conversion_rate_national_currency_per_current_international_dollar	0.0286
35	current_health_expenditure_(%_of_gdp)	-0.0245
36	volume_of_imports_of_goods_and_services_percent_change	-0.0243
37	gross_domestic_product,_current_prices_u.s._dollars	-0.0229
38	inflation,_average_consumer_prices_index	-0.0206
39	tax_revenue_(%_of_gdp)	0.0200
40	political_stability_and_absence_of_violence/terrorism:_estimate	-0.0174
41	current_health_expenditure_per_capita,_ppp_(current_international_\$)	-0.0149
42	trade_in_services_(%_of_gdp)	0.0147
43	government_effectiveness:_estimate	0.0123
44	goods_imports_(bop,_current_us\$)	-0.0119
45	general_government_primary_net_lending/borrowing_national_currency	0.0110

Annex - B

Model Classes	Normalized	Moody's	S&P	Fitch	DBRS	Investment Category
20	29	Aaa	AAA	AAA	AAA	Investment Grade: Prime / Highest Quality
19	28	Aa1	AA+	AA+	AA (high)	Investment Grade: High Quality
18	27	Aa2	AA	AA	AA	
17	26	Aa3	AA-	AA-	AA (low)	
16	25	A1	A+	A+	A (high)	Investment Grade: Upper Medium Quality
15	24	A2	A	A	A	
14	23	A3	A-	A-	A (low)	
13	22	Baa1	BBB+	BBB+	BBB (high)	
12	21	Baa2	BBB	BBB	BBB	
11	20	Baa3	BBB-	BBB-	BBB (low)	Investment Grade: Lowest Tier
10	15	Ba1	BB+	BB+	BB (high)	Speculative Grade (Junk): Highest Quality Junk
9	14	Ba2	BB	BB	BB	Speculative Grade (Junk)
8	13	Ba3	BB-	BB-	BB (low)	
7	12	B1	B+	B+	B (high)	Highly Speculative
6	11	B2	B	B	B	
5	10	B3	B-	B-	B (low)	
4	8	Caa1	CCC+	CCC+	CCC (high)	Substantial Risk / Very High Credit Risk
3	7	Caa2	CCC+	CCC+	CCC	Substantial Risk
2	6	Caa3	CCC-	CCC-	CCC (low)	Extremely Speculative
1	5	Ca	CC	CC	CC	Near Default
0	1	C	C	C	C	Default Imminent / In Default